SSBMS-SLM

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Solid-State Lithium-Metal Battery Management System

Automotive EV Battery Electronics | Rev A2 - Architecture, Atomic Requirements, Interfaces & Verification Baseline

Mission profile. The product supervises solid-state lithium-metal cells whose electrical behavior, interfacial impedance and mechanical expansion/pressure characteristics require coordinated observability. The BMS measures cell voltage, temperature, pack current and expansion/pressure, estimates SOC/SOH using a non-linear Kalman framework, manages contactors and active balancing, and provides vehicle-level status while preventing invalid data or communication services from increasing cell stress.

Engineering baseline. The architecture uses high-impedance cell-monitor AFE devices, isolated daisy-chain communication, pressure/expansion sensing, pack current measurement, high-voltage isolation and contactor/precharge supervision. Active balancing capability is controlled up to the source-defined 5 A per cell, subject to released thermal and cell-energy limits. AES-128 protection is retained for the isolated monitor network. Values not fixed by the product baseline remain configuration-controlled parameters and require objective release evidence rather than assumption.

Definition	Controlled product interpretation
Product class	Automotive BMS optimized for solid-state lithium-metal cell packs with electrical, thermal and expansion/pressure supervision.
Safety / assurance	ASIL-C safety intent for voltage and expansion-pressure monitoring. Final ISO 26262 HARA/FSC/TSC allocation, ASIL decomposition and compliance evidence remain program-specific.
Lifecycle mission	15-year vehicle lifecycle with impedance-growth, pressure-sensor drift, balancing usage, contactor events, thermal exposure and NVM health tracked for predictive maintenance.
Primary architecture	High-impedance AFE chain; cell voltage/temperature/pressure sensing; pack current and isolation monitoring; contactor/precharge control; active balancing up to 5 A/cell; non-linear Kalman SOC/SOH; isolated AES-128 daisy chain.
Release philosophy	Hardware-independent safety limits are separated from supplier, cell chemistry, pack mechanics, calibration and installation parameters. Evidence is configuration specific and change controlled.

Release interpretation. No planning value, typical supplier datasheet value, unverified cell limit or regional assumption is a released requirement unless it is configuration controlled and supported by approved analysis, test or inspection evidence.

PRODUCT GOVERNANCE, REQUIREMENT QUALITY & CHANGE CONTROL

Single-intent requirements, controlled cell/pack parameters, explicit ownership and evidence-based release.

The BMS separates immutable safety limits from cell-supplier characteristics, pack-mechanical constraints, calibrations and installation-dependent values. Fast protection mechanisms are allocated by physical response time and independence; supervisory software owns classification, logging, derating and controlled recovery. A change to cell chemistry, AFE, pressure sensor, balancing topology, contactor, estimation model or pack pressure structure may reuse prior evidence only when similarity analysis proves that the affected stress, uncertainty and safety assumptions remain bounded.

Class	Examples	Release rule
Immutable safety	Absolute OV/UV, overcurrent, temperature/pressure hard limits, contactor inhibit paths	Change requires safety impact analysis and affected DVP regression.
Variant controlled	Cell family, pack series/parallel count, balancing hardware, AFE population, contactor set	Released by signed variant coding, BOM and calibration package.
Supplier dependent	AFE error, pressure-sensor drift, shunt/Hall accuracy, insulation, cell impedance/expansion data	Use guaranteed production data and WCCA; requalify on source change.
Installation dependent	Pack pressure stack-up, busbar/harness impedance, cooling interface, grounding, isolation path	Captured in pack ICD and validated at integration.
Calibration	Estimator parameters, thresholds, derate curves, timing, filters and model maps	Authenticated, range checked and traceable to verification evidence.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-GOV-001	Each product requirement shall trace upward to a mission need, hazard/risk control or external interface obligation and downward to at least one objective verification artifact tied to hardware revision, software build, calibration set and cell/pack variant.	Bidirectional requirements and DVP traceability audit.
SSBMS-GOV-002	A supplier, topology, algorithm, cell chemistry or mechanical-pack change shall not reuse prior evidence until documented impact analysis confirms that measurement uncertainty, electrical/thermal stress, pressure behavior and safety independence remain inside approved similarity bounds.	Change-control and similarity review.
SSBMS-GOV-003	Safety-relevant calibrations and model parameters shall be authenticated, range checked, revision controlled and recoverable; production software shall not silently substitute an unapproved default that increases operating permissiveness.	Calibration integrity and rollback test.
SSBMS-GOV-004	Every TBC/TBD affecting safety, cell limits, measurement accuracy, timing, balancing energy, thermal behavior or release acceptance shall have an owner, closure method, due gate and linked evidence before production release.	Parameter/TBC register closure audit.

Verification, margin and future-change rule

Requirement wording is single intent so one test, analysis or inspection can determine compliance. Global verification boundary: unless a section defines a narrower envelope, WCCA and verification use the released beginning- and end-of-life bounds for sensor error, cell impedance, temperature, pressure/preload, supply tolerance and mechanical stack-up. Global release evidence records the cell family, hardware revision, software build, calibration set, supplier-dependent limits, measurement uncertainty, environmental/test conditions and retained raw data. Subsystem change control therefore reopens only the analyses and tests whose physical, algorithmic, process or interface assumptions are changed.

VEHICLE / PACK ITEM BOUNDARY & COMMAND AUTHORITY

The BMS is one element of the traction-energy function; vehicle torque, charging and thermal controllers depend on its limits but cannot override local cell protection.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P03-01	Every released interface for vehicle/pack functional boundary, command ownership and external mitigation dependencies shall define source, destination, units, encoding, valid range, update rate, freshness/integrity status, timeout and authority; ambiguous ownership is not permitted.	ICD and safety-allocation review.
SSBMS-P03-02	charge/discharge/current/contactor authority arbitration shall remain deterministic and bounded during message loss, duplication, corruption, bus recovery, reset and maximum specified network or daisy-chain loading.	HIL authority-conflict and timeout matrix.
SSBMS-P03-03	Invalid, stale or unauthenticated information associated with external command or status dependency shall be treated as non-permissive and shall not override autonomous local voltage, current, thermal, pressure or contactor protection.	Vehicle-command loss/corruption fault injection.

Verification, margin and future-change rule

The BMS is one element of the traction-energy function; vehicle torque, charging and thermal controllers depend on its limits but cannot override local cell protection. Data that is late, duplicated, corrupted or ambiguously owned can be more hazardous than a clean communication loss. Interface changes require protocol, authority, freshness, timeout and non-permissive fallback regression; local electrical, thermal, pressure and contactor protections remain autonomous.

MISSION PROFILE, OPERATING MODES & DEGRADED STATES

The released mission profile covers sleep, wake, precharge, drive, regenerative charging, plug-in charging, balancing, service, storage and faulted states across the approved cell and pack variants.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P04-01	The mode-control baseline shall define each operating state, legal transition, entry/exit criterion, command owner and permitted energy flow, including the default response to invalid or contradictory state inputs.	State-machine and mission-profile review.
SSBMS-P04-02	mode transitions and degraded-state capability shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	HIL transition sweep with brownout and communication faults.
SSBMS-P04-03	Loss, invalidity or disagreement affecting mode authority, readiness or recovery information shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Invalid-state/recovery injection.

Verification, margin and future-change rule

The released mission profile covers sleep, wake, precharge, drive, regenerative charging, plug-in charging, balancing, service, storage and faulted states across approved cell and pack variants. Mode-transition verification shall exercise interrupted wake-up, aborted precharge, charger handover, low-energy balancing, brownout and service entry/exit behavior. Changes to mode authority, contactor sequencing, wake sources, charger interaction or vehicle-state dependencies require targeted state-machine and fault-recovery regression.

SOLID-STATE CELL / PACK ELECTRICAL-MECHANICAL ARCHITECTURE

Solid-state lithium-metal cells couple electrical state, interfacial impedance, temperature and mechanical expansion; the pack architecture must make those interactions observable without assuming a conventional liquid-electrolyte cell model.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P05-01	The pack architecture shall define cell/module segmentation, voltage/temperature/pressure sensing locations, electrical isolation boundaries and the mechanical preload interfaces that affect measurement validity.	Released block/pack diagram review.
SSBMS-P05-02	electrical and mechanical observability shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Representative module correlation and tolerance analysis.
SSBMS-P05-03	Loss, invalidity or disagreement affecting cell-stack topology or pressure-boundary information shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Variant mismatch and missing-boundary test.

Verification, margin and future-change rule

Solid-state lithium-metal cells couple electrical state, interfacial impedance, temperature and mechanical expansion; the pack architecture must make those interactions observable without assuming a conventional liquid-electrolyte cell model. Cell format, module compression strategy, busbar geometry, pressure-transfer path or cell-supplier mechanical limits that change the coupled plant require renewed electrical-mechanical correlation and pack-interface review.

HIGH-IMPEDANCE AFE & CELL-MONITOR ARCHITECTURE

Cell-monitor channels require high input impedance, controlled leakage and error budgeting so the measurement system does not materially load high-impedance solid-state cells or hide open-wire conditions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P06-01	The AFE architecture shall define channel allocation, input impedance/leakage limits, protection network, ADC/reference path, open-wire strategy and the diagnostics used to declare a cell-voltage sample valid.	AFE WCCA and schematic review.
SSBMS-P06-02	cell-voltage acquisition accuracy and channel integrity shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Calibrated cell-simulator sweep plus open-wire injection.
SSBMS-P06-03	Loss, invalidity or disagreement affecting AFE channel validity, reference or supply health shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	AFE supply/reference/channel fault injection.

Verification, margin and future-change rule

Cell-monitor channels require high input impedance, controlled leakage and error budgeting so the measurement system does not materially load high-impedance solid-state cells or hide open-wire conditions. An AFE IC, input filter, cell-protection network, ADC reference, connector leakage path or open-wire injection topology change requires repeat leakage/error WCCA and representative open-wire discrimination testing.

EXPANSION / PRESSURE SENSING & PLAUSIBILITY

Mechanical expansion/pressure is treated as a first-class pack state variable. Sensor range, mounting transfer path, temperature effects and structural preload are part of the released measurement chain.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P07-01	The pressure/expansion measurement architecture shall define sensor location, preload transfer path, calibrated range, temperature compensation, plausibility relationships and ownership of each derived pressure state.	Mechanical/sensor ICD and error-budget review.
SSBMS-P07-02	pressure trend, dynamic plausibility and cross-channel agreement shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Pressure-rig sweep across temperature and preload.
SSBMS-P07-03	Loss, invalidity or disagreement affecting pressure sensor validity or structural transfer-path data shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Frozen/offset/disagreement injection.

Verification, margin and future-change rule

Mechanical expansion/pressure is treated as a first-class pack state variable. Pressure-sensor technology, mounting stack, preload, adhesive/flex construction, load-spreader geometry or pressure-transfer-path changes require renewed mechanical correlation, calibration and cross-channel plausibility verification across temperature and lifecycle conditions.

PACK CURRENT SENSING, COULOMB COUNTING & DYNAMIC RANGE

Pack current supports SOC propagation, power limiting, fault detection and contactor diagnostics. The measurement chain must remain useful from sleep/leakage observation through propulsion and fast-charge transients.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P08-01	The pack-current measurement chain shall define sensing technology, bidirectional range, offset strategy, bandwidth, saturation behavior, calibration source and the consumers allowed to use each accuracy class.	Current metrology WCCA.
SSBMS-P08-02	current measurement and coulomb-count integration shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Bidirectional current sweep and pulse correlation.
SSBMS-P08-03	Loss, invalidity or disagreement affecting current validity, offset correction or saturation status shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Saturation/frozen/offset fault injection.

Verification, margin and future-change rule

Pack current supports SOC propagation, power limiting, fault detection and contactor diagnostics. A change from shunt to magnetic sensing, range architecture, busbar geometry, ADC path, offset compensation, filtering or calibration strategy requires repeat bidirectional metrology, saturation recovery and coulomb-count correlation.

CONTACTOR, PRECHARGE, DISCHARGE & HV SEQUENCING

The BMS owns the traction-battery energization sequence within the vehicle architecture. Precharge completion, contactor feedback, isolation status and voltage plausibility must be qualified before high-energy connection.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P09-01	The safety concept shall trace contactor/precharge/discharge topology, feedback and state authority from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	Power-path schematic/FMEA review.
SSBMS-P09-02	precharge timing, weld detection and bus-voltage plausibility shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	Power-HIL precharge and weld-fault matrix.
SSBMS-P09-03	A fault affecting contactor feedback, HV voltage or isolation readiness shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	Feedback mismatch and brownout injection.

Verification, margin and future-change rule

The BMS owns the traction-battery energization sequence within the vehicle architecture. The safety argument includes physical detection latency, signal conditioning, propagation, decision, actuation and energy decay rather than software task time alone. Any change to cell limits, sensor technology, contactor path, balancing hardware or pack mechanics reopens the linked FTTI and fault-containment evidence.

SOC / SOH ESTIMATION FRAMEWORK

The non-linear Kalman estimation framework combines current integration with voltage, temperature, impedance and pressure-related information while declaring confidence and model applicability.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P10-01	The released estimation function shall configuration-control Kalman model structure, state vector, covariance limits and calibration ownership, including model version, state variables, calibration set, numerical limits, data quality rules and applicability to each cell/pack variant.	Model/parameter configuration review.
SSBMS-P10-02	SOC/SOH convergence and bounded estimation error shall meet the released accuracy, convergence and bounded-error criteria over representative temperature, SOC, SOH, pressure, impedance and duty-cycle conditions; extrapolation beyond validated regions shall be explicitly limited.	Cell/pack replay against reference instrumentation.
SSBMS-P10-03	When model confidence, covariance or required sensor data is unavailable, stale or outside the validated model domain, the estimator shall declare reduced confidence and shall not silently increase charge, discharge, balancing or contactor permissiveness.	Sensor dropout and out-of-domain model test.

Verification, margin and future-change rule

The non-linear Kalman estimation framework combines current integration with voltage, temperature, impedance and pressure-related information while declaring confidence and model applicability. Model validity is bounded by the characterized cell chemistry, temperature, pressure, impedance and duty-cycle domain. Parameter, estimator, sensor or cell-supplier changes require model-identification impact analysis, reference-data comparison and regression of any limit or diagnostic that consumes the estimator output.

ACTIVE BALANCING ARCHITECTURE & CELL ENERGY CONTROL

Active balancing up to 5 A per cell is an energy-transfer function, not only a SOC equalization feature. Switch states, current measurement, isolation and thermal interaction are controlled at cell level.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P11-01	The safety concept shall trace balancing topology, current paths, switch states and per-cell authorization from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	Balancing schematic/SOA review.
SSBMS-P11-02	balancing current, energy transfer and thermal limits shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	Cell-simulator and calorimetric balancing test.
SSBMS-P11-03	A fault affecting balancing feedback, destination/source cell validity or temperature data shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	Switch-stuck/sensor-loss/thermal fault injection.

Verification, margin and future-change rule

Active balancing up to 5 A per cell is an energy-transfer function, not only a SOC equalization feature. The safety argument includes physical detection latency, signal conditioning, propagation, decision, actuation and energy decay rather than software task time alone. Any change to cell limits, sensor technology, contactor path, balancing hardware or pack mechanics reopens the linked FTTI and fault-containment evidence.

SOFTWARE ARCHITECTURE, STATE MACHINE & DIAGNOSTIC OWNERSHIP

Software separates acquisition, estimation, protection supervision, balancing, contactor control, communications and lifecycle logging with deterministic scheduling and explicit authority.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P12-01	The software architecture shall define acquisition, estimation, protection, balancing, contactor, communications and lifecycle partitions together with task rates, startup order, inter-task data ownership and recovery authority.	Software architecture and timing review.
SSBMS-P12-02	scheduler timing, data freshness and diagnostic execution shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	SIL/HIL CPU-load and fault-burst stress.
SSBMS-P12-03	Loss, invalidity or disagreement affecting task execution, watchdog milestone or data-validity status shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Task-stall/stale-data/watchdog injection.

Verification, margin and future-change rule

Software separates acquisition, estimation, protection supervision, balancing, contactor control, communications and lifecycle logging with deterministic scheduling and explicit authority. Scheduler, RTOS/BSW, task-rate, memory-map, compiler or inter-task data-contract changes require timing/WCET, startup-order and invalid-data regression for the affected software safety mechanisms.

ISO 26262 SAFETY INTENT & PROGRAM-SPECIFIC ALLOCATION

The product baseline carries ASIL-C safety intent for voltage and expansion-pressure monitoring. Final hazard classification, ASIL allocation, decomposition and quantitative metrics remain project-specific.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P13-01	The safety concept shall trace ASIL-C intent, HARA/FSC/TSC assumptions and safety-mechanism allocation from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	Safety plan/HARA/FSC/TSC trace review.
SSBMS-P13-02	FTTI, diagnostic coverage and independence evidence shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	FMEDA plus fault-injection timing evidence.
SSBMS-P13-03	A fault affecting safety mechanism, common-cause assumption or external mitigation shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	Common-cause and mechanism-unavailable tests.

Verification, margin and future-change rule

The product baseline carries ASIL-C safety intent for voltage and expansion-pressure monitoring. The safety argument includes physical detection latency, signal conditioning, propagation, decision, actuation and energy decay rather than software task time alone. Any change to cell limits, sensor technology, contactor path, balancing hardware or pack mechanics reopens the linked FTTI and fault-containment evidence.

POWER, THERMAL & LOW-VOLTAGE SUPPLY ARCHITECTURE

The BMS low-voltage supply, AFE bias, isolation power and balancing losses must remain bounded across vehicle supply transients, hot soak, cold start and degraded cooling.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P14-01	The low-voltage power architecture shall define the PMIC/regulator tree, AFE and isolation supplies, brownout/reset thresholds, balancing-loss paths, temperature sensing and the safe default of every energy-control output.	Power-tree WCCA and thermal model review.
SSBMS-P14-02	supply sequencing, thermal margin and restart behavior shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Supply-transient plus chamber testing.
SSBMS-P14-03	Loss, invalidity or disagreement affecting temperature, supply or cooling-interface validity shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Brownout/hot-soak/sensor invalid injection.

Verification, margin and future-change rule

The BMS low-voltage supply, AFE bias, isolation power and balancing losses must remain bounded across vehicle supply transients, hot soak, cold start and degraded cooling. PMIC, isolated-bias, regulator, decoupling, balancing-switch or thermal-interface changes require renewed power-tree transient analysis, brownout sequencing checks and thermal-loss correlation.

ISOLATED DAISY-CHAIN COMMUNICATION & AES-128 PROTECTION

Module-monitor communication is isolated and encrypted. Physical-layer integrity, retry behavior, message ordering and cryptographic protection are all subordinate to local cell-safety limits.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P15-01	Every released interface for isolated daisy-chain topology, addressing, framing and AES-128 key ownership shall define source, destination, units, encoding, valid range, update rate, freshness/integrity status, timeout and authority; ambiguous ownership is not permitted.	Network/security architecture review.
SSBMS-P15-02	communication latency, error recovery and message integrity shall remain deterministic and bounded during message loss, duplication, corruption, bus recovery, reset and maximum specified network or daisy-chain loading.	Error-rate, retry and cryptographic negative test.
SSBMS-P15-03	Invalid, stale or unauthenticated information associated with daisy-chain freshness, authentication or isolation status shall be treated as non-permissive and shall not override autonomous local voltage, current, thermal, pressure or contactor protection.	Link-break/replay/corruption injection.

Verification, margin and future-change rule

Module-monitor communication is isolated and encrypted. Data that is late, duplicated, corrupted or ambiguously owned can be more hazardous than a clean communication loss. Interface changes require protocol, authority, freshness, timeout and non-permissive fallback regression; local electrical, thermal, pressure and contactor protections remain autonomous.

MECHANICAL INTEGRATION INTO PACK PRESSURE-CONTAINMENT STRUCTURE

The BMS sensing hardware is integrated with a pressure-bearing pack environment. PCB, sensor, connector and flex interconnect placement must not invalidate the pack preload or distort pressure measurements.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P16-01	Mechanical variant approval shall bind PCB/sensor mounting, pressure-transfer path, connector/flex routing, service stack-up and structural limits to a configuration-controlled pack drawing and assembly process.	Mechanical stack-up and tolerance review.
SSBMS-P16-02	mechanical stack-up, strain transfer and sensing repeatability shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Pressure/strain correlation after vibration and thermal cycle.
SSBMS-P16-03	If evidence for mounting/preload/stack-up configuration data is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Mounting-variation and connector-stress test.

Verification, margin and future-change rule

The BMS sensing hardware is integrated with a pressure-bearing pack environment. Mechanical release evidence shall correlate sensor/PCB/flex placement, preload transfer and connector retention to the serialized pack stack-up and service assembly process. Cell format, enclosure, flex routing, sensor mount or service-clamp changes require mechanical delta analysis and targeted pressure/strain regression before variant approval.

MANUFACTURING, EOL PRESSURE CALIBRATION & OFFSET LOGGING

Production must demonstrate pressure-sensor calibration, electrical channel integrity, isolation, contactor I/O and serialized configuration before a pack or BMS variant is released.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P17-01	EOL calibration release shall define fixture identity, reference standards, applied preload, calibration temperature, measurement uncertainty, stored offsets and serialized linkage to the tested BMS/pack unit.	Manufacturing/EOL control-plan review.
SSBMS-P17-02	calibration repeatability, channel coverage and configuration integrity shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Gauge R&R and golden-unit correlation.
SSBMS-P17-03	If evidence for EOL pressure calibration and serialized acceptance evidence is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Missing/invalid EOL evidence blocks release.

Verification, margin and future-change rule

Production must demonstrate pressure-sensor calibration, electrical channel integrity, isolation, contactor I/O and serialized configuration before a pack or BMS variant is released. EOL evidence shall include traceable pressure references, gauge capability, calibration-temperature/preload conditions, programmed software/calibration identity and retained pass/fail records. Fixture, reference standard, preload tooling or calibration algorithm changes require EOL correlation before release.

15-YEAR LIFECYCLE, IMPEDANCE GROWTH & PREDICTIVE MAINTENANCE

The lifecycle strategy tracks cell impedance growth, pressure evolution, thermal exposure, balancing duty, contactor events and sensor drift so service decisions are based on trend evidence rather than a single threshold.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P18-01	The released estimation function shall configuration-control lifecycle counters, impedance/pressure history, NVM endurance and service thresholds, including model version, state variables, calibration set, numerical limits, data quality rules and applicability to each cell/pack variant.	Lifecycle model and NVM budget review.
SSBMS-P18-02	aging trend quality and predictive-maintenance confidence shall meet the released accuracy, convergence and bounded-error criteria over representative temperature, SOC, SOH, pressure, impedance and duty-cycle conditions; extrapolation beyond validated regions shall be explicitly limited.	Accelerated-aging plus field-data replay.
SSBMS-P18-03	When aging model, counter integrity or retained history is unavailable, stale or outside the validated model domain, the estimator shall declare reduced confidence and shall not silently increase charge, discharge, balancing or contactor permissiveness.	Counter corruption/model invalidity test.

Verification, margin and future-change rule

The lifecycle strategy tracks cell impedance growth, pressure evolution, thermal exposure, balancing duty, contactor events and sensor drift so service decisions are based on trend evidence rather than a single threshold. Model validity is bounded by the characterized cell chemistry, temperature, pressure, impedance and duty-cycle domain. Parameter, estimator, sensor or cell-supplier changes require model-identification impact analysis, reference-data comparison and regression of any limit or diagnostic that consumes the estimator output.

PRODUCT-LEVEL DVP, HIL & FAULT-INJECTION STRATEGY

Verification combines cell simulators, pressure rigs, power-HIL, pack-HIL, thermal chambers, EMC testing and representative hardware so combined electrical, mechanical and timing faults are exercised.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P19-01	The product DVP baseline shall define representative operating corners, fault cases, instrumentation, plant/load models, raw-data retention and the hardware/software/calibration identity associated with each pass/fail result.	Configuration-controlled DVP review.
SSBMS-P19-02	test coverage of operating corners, single faults and combined stresses shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Calibrated test execution with raw data.
SSBMS-P19-03	If evidence for verification evidence or released pass/fail criteria is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Open critical evidence prevents release.

Verification, margin and future-change rule

Verification combines cell simulators, pressure rigs, power-HIL, pack-HIL, thermal chambers, EMC testing and representative hardware so combined electrical, mechanical and timing faults are exercised. The DVP shall identify the exact plant/load models, fault matrix, instrumentation accuracy, hardware/software/calibration identity and raw-data retention policy for every release claim. A changed operating envelope, cell family, fault model or test setup reopens only the affected DVP rows after documented impact review.

INTERFACE DATA CONTRACTS & VEHICLE INTEGRATION

Vehicle and charger interfaces consume BMS limits, SOC/SOH, isolation, thermal and pressure status. Every permissive decision must be based on fresh, plausible and authorized data with defined timeout behavior.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P20-01	Every released interface for vehicle/charger/thermal interface signals, authority and timing shall define source, destination, units, encoding, valid range, update rate, freshness/integrity status, timeout and authority; ambiguous ownership is not permitted.	ICD/data dictionary review.
SSBMS-P20-02	data-contract freshness, range and fallback behavior shall remain deterministic and bounded during message loss, duplication, corruption, bus recovery, reset and maximum specified network or daisy-chain loading.	CAN FD/HIL timeout and range injection.
SSBMS-P20-03	Invalid, stale or unauthenticated information associated with charge/discharge limit, SOC/SOH or fault-status validity shall be treated as non-permissive and shall not override autonomous local voltage, current, thermal, pressure or contactor protection.	Stale/implausible command and status test.

Verification, margin and future-change rule

Vehicle and charger interfaces consume BMS limits, SOC/SOH, isolation, thermal and pressure status. Data that is late, duplicated, corrupted or ambiguously owned can be more hazardous than a clean communication loss. Interface changes require protocol, authority, freshness, timeout and non-permissive fallback regression; local electrical, thermal, pressure and contactor protections remain autonomous.

PART I RELEASE GATES & CONTROLLED PARAMETER REGISTER

System-level release requires closure of item boundaries, cell/pack variants, safety allocation, measurement budgets, model domains, thermal/mechanical assumptions, EOL criteria and objective verification evidence.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-P21-01	The controlled parameter register shall assign an owner, source, status, closure criterion, evidence link and affected variant to every TBC/TBD value that can influence safety, performance, manufacturing or service behavior.	Release-readiness review.
SSBMS-P21-02	cross-document consistency among PRD, ICD, safety, DVP, BOM and calibration shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Traceability and configuration audit.
SSBMS-P21-03	If evidence for unclosed safety, model, interface, manufacturing or lifecycle parameter is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Any unresolved high-severity or unbounded parameter blocks release.

Verification, margin and future-change rule

System-level release requires closure of item boundaries, cell/pack variants, safety allocation, measurement budgets, model domains, thermal/mechanical assumptions, EOL criteria and objective verification evidence. The release record shall link each open/closed parameter to its owner, evidence artifact, affected requirement and approved hardware/software/cell variant. Conflicting ownership, unresolved high-severity mechanisms or unverified derived margins remain explicit release blockers.

PART II - DETAILED SUBSYSTEM PRDs & ENGINEERING ASSURANCE ANNEX

Expanded architecture, atomic requirements, fault containment, interfaces, DVP and lifecycle evidence for the solid-state BMS.

This annex decomposes the product-level baseline into detailed subsystem requirements for electrical measurement, expansion/pressure observability, high-voltage sequencing, active balancing, estimation, communications, cybersecurity, thermal/mechanical integration, manufacturing and lifecycle assurance. Each annex page defines the mechanism that must be bounded, the single-intent requirements that establish ownership and behavior, and the objective evidence required for release.

Controlled parameters are used wherever cell supplier data, pack series/parallel configuration, pressure preload, current-sensor range, AFE performance, isolation network, contactor selection, thermal interface, software timing or vehicle integration has not fixed a universal value. A parameter may be supplier dependent or installation dependent without becoming optional: it must still have an owner, an approved range, a verification method and a traceable configuration identity.

Across Part II, runtime faults are deliberately separated from engineering evidence gaps. Invalid sensor data, communication loss or detected electrical faults may require a non-permissive or derated operating state. By contrast, missing qualification, manufacturing, DVP, lifecycle or supplier-change evidence is a release blocker for the affected hardware/BOM/installation variant; it is not represented as a fictitious runtime BMS reaction. This distinction preserves technically correct allocation and prevents template language from replacing engineering judgment.

Evidence shall retain raw records and identify cell lot or emulator configuration, hardware revision, software build, calibration set, environmental condition, pressure/mechanical fixture, instrument calibration and any external model used to declare a pass. Prior evidence is reusable only after documented similarity analysis demonstrates that electrical, thermal, mechanical, timing, cyber and safety margins remain inside the assumptions of the original evidence.

PACK POWER-ON / POWER-OFF & CONTACTOR STATE MACHINE

Detailed subsystem PRD / engineering assurance annex - Incorrect sequencing can connect energy before voltage, isolation, contactor feedback, pressure/temperature readiness or controller health is qualified.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A01-01	The safety concept shall trace state authority, precharge/contactors, readiness conditions and recovery timing from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	State-machine/ICD review.
SSBMS-A01-02	state transitions, contactor sequencing and brownout recovery shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	Power-HIL transition sweep with weld/open/precharge faults.
SSBMS-A01-03	A fault affecting voltage/isolation/contactor feedback or readiness data shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	Invalid-feedback and brownout injection.

Verification, margin and future-change rule

Incorrect sequencing can connect energy before voltage, isolation, contactor feedback, pressure/temperature readiness or controller health is qualified. Power-on timing shall be measured from wake request through sensor validation, precharge decision, contactor actuation and DC-link settling, including bounce and brownout recovery. Changes to wake logic, contactor hardware, bus sensing, precharge components or readiness criteria require end-to-end sequencing regression.

AFE ERROR BUDGET, LEAKAGE & OPEN-WIRE DETECTION

Detailed subsystem PRD / engineering assurance annex - High-impedance solid-state cell measurement is sensitive to input leakage, bias currents, protection networks, reference error and open-wire behavior.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A02-01	AFE measurement release shall bind leakage, gain/offset/reference error, input protection, channel mapping and open-wire behavior to the selected device, PCB network, harness leakage and cell impedance envelope.	AFE WCCA and schematic review.
SSBMS-A02-02	cell-voltage accuracy and open-wire discrimination shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Cell-simulator sweep plus open-wire matrix.
SSBMS-A02-03	Loss, invalidity or disagreement affecting channel/reference/supply validity shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Reference/channel fault injection.

Verification, margin and future-change rule

High-impedance solid-state cell measurement is sensitive to input leakage, bias currents, protection networks, reference error and open-wire behavior. Cell-voltage accuracy and open-wire discrimination shall be correlated to representative hardware across the released impedance and temperature envelope. AFE IC, reference, input RC/protection or harness leakage changes reopen the channel-error and open-wire WCCA.

PRESSURE SENSOR METROLOGY, PRELOAD & DRIFT

Detailed subsystem PRD / engineering assurance annex - Pressure observability depends on sensor error and the mechanical transfer path; a numerically valid sensor can still be wrong if preload or stack-up changes.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A03-01	Pressure-sensing release shall bind sensor range, preload transfer function, temperature coefficient, mounting stack and calibration coefficients to the representative pack mechanical configuration.	Mechanical/sensor error-budget review.
SSBMS-A03-02	pressure accuracy, dynamic response and cross-channel plausibility shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Pressure-rig sweep over temperature/preload.
SSBMS-A03-03	Loss, invalidity or disagreement affecting pressure sensor or preload/stack-up validity shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Drift/frozen/offset/stack-up fault injection.

Verification, margin and future-change rule

Pressure observability depends on both sensor accuracy and the mechanical transfer path; a numerically valid channel can still be wrong when preload or stack-up changes. Pressure accuracy, dynamic response and cross-channel plausibility shall be correlated on a representative pressure rig. Sensor technology, mounting material, preload process, adhesive/flex or pack-load-path changes require renewed calibration and structural correlation.

PACK CURRENT METROLOGY & COULOMB-COUNT ACCURACY

Detailed subsystem PRD / engineering assurance annex - Current offset and saturation directly influence SOC, power limits, fault detection and contactor diagnostics, so the complete bidirectional error budget is configuration controlled.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A04-01	Current-metrology release shall bind sensing range, offset, bandwidth, temperature drift, ADC path and calibration coefficients to the busbar/shunt or magnetic-sensor implementation used in the pack.	Current WCCA and metrology review.
SSBMS-A04-02	bidirectional current accuracy and charge integration shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Calibrated bidirectional pulse/sweep test.
SSBMS-A04-03	Loss, invalidity or disagreement affecting current saturation/offset correction/validity shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Offset/saturation/frozen sensor injection.

Verification, margin and future-change rule

Current offset and saturation directly influence SOC, power limits, fault detection and contactor diagnostics, so the complete bidirectional error budget is configuration controlled. Bidirectional current accuracy, zero-current offset recovery and charge integration shall be correlated to representative hardware. Shunt/Hall technology, range, amplifier/ADC path, busbar layout or offset-compensation changes reopen metrology and coulomb-count validation.

PRECHARGE, CONTACTOR WELD & FEEDBACK DIAGNOSTICS

Detailed subsystem PRD / engineering assurance annex - The BMS must distinguish a legitimate slow precharge from welded contacts, open contactors, incorrect bus feedback or abnormal load capacitance without destructive retry cycling.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A05-01	The safety concept shall trace precharge resistance/capacitance assumptions, contactor feedback and weld-detection thresholds from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	Power-path FMEA and threshold review.
SSBMS-A05-02	precharge completion/weld/open-contact discrimination shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	Power-HIL capacitance/source sweep.
SSBMS-A05-03	A fault affecting bus-voltage or contactor-feedback validity shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	Weld/open/feedback mismatch fault injection.

Verification, margin and future-change rule

The BMS must distinguish a legitimate slow precharge from welded contacts, open contactors, incorrect bus feedback or abnormal load capacitance without destructive retry cycling. Precharge verification shall cover RC tolerance, source/load impedance, feedback delay, contact bounce and timeout classification from command to settled bus voltage. Precharge resistor/capacitance, contactor, voltage-sense or retry-policy changes reopen weld/open discrimination tests.

HV ISOLATION MONITORING & LEAKAGE PLAUSIBILITY

Detailed subsystem PRD / engineering assurance annex - Isolation status constrains contactor closure and vehicle operation; monitor range, response time, switching interference and external charger/inverter coupling are part of the item boundary.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A06-01	The safety concept shall trace isolation-monitor topology, excitation, thresholds and external coupling assumptions from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	Isolation architecture and WCCA review.
SSBMS-A06-02	isolation-fault detection and false-trip immunity shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	Programmable isolation-resistance sweep.
SSBMS-A06-03	A fault affecting isolation data or monitor health shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	Monitor-stuck/interference/fault injection.

Verification, margin and future-change rule

Isolation status constrains contactor closure and vehicle operation; monitor range, response time, switching interference and external charger/inverter coupling are part of the item boundary. Isolation-fault timing shall include monitor excitation, filtering, threshold persistence, communication latency where applicable and contactor-energy removal. IMD topology, HV filtering, charger/inverter coupling or isolation-threshold changes require repeat detection and false-trip immunity testing.

FAST CELL OV / UV PROTECTION & MEASUREMENT INDEPENDENCE

Detailed subsystem PRD / engineering assurance annex - Cell overvoltage and undervoltage protection must remain credible when the application processor stalls or estimation software is unavailable.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A07-01	The safety concept shall trace OV/UV thresholds, comparator/AFE path independence and gate/contactor inhibit ownership from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	Safety architecture/FMEDA review.
SSBMS-A07-02	fault-onset-to-energy-reduction timing and diagnostic coverage shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	Boundary voltage sweep with processor stall.
SSBMS-A07-03	A fault affecting cell-voltage protection path or AFE validity shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	AFE/comparator/processor fault injection.

Verification, margin and future-change rule

Cell overvoltage and undervoltage protection must remain credible when the application processor stalls or estimation software is unavailable. Fault-onset-to-energy-reduction testing shall capture cell/AFE acquisition, independent threshold detection, propagation to the inhibit path and contactor or charger/inverter response. AFE, comparator path, contactor interface or threshold-filter changes reopen protection timing and diagnostic-coverage evidence.

OVERTEMPERATURE, PRESSURE & RUNAWAY-PRECONDITION DETECTION

Detailed subsystem PRD / engineering assurance annex - Solid-state cell safety supervision combines temperature and expansion/pressure trends so abnormal mechanical/electrochemical behavior is detected without relying on a single scalar limit.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A08-01	The safety concept shall trace temperature/pressure thresholds, rate-of-change features, plausibility and independent reaction paths from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	Safety concept and threshold rationale review.
SSBMS-A08-02	precondition detection, false-positive control and FTTI shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	Thermal/pressure rig abnormality injection.
SSBMS-A08-03	A fault affecting temperature/pressure sensing or cross-plausibility shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	Sensor loss/disagreement/common-cause test.

Verification, margin and future-change rule

Solid-state cell safety supervision combines temperature and expansion/pressure trends so abnormal mechanical-electrochemical behavior is detected without relying on a single scalar limit. Detection verification shall cover sensor dynamics, trend windows, persistence logic, cross-channel plausibility and the physical delay to charge/discharge-energy reduction. Pressure sensor, thermal sensor, feature extraction or persistence-threshold changes require renewed false-positive and FTTI testing.

ACTIVE BALANCING SWITCHING, CURRENT CONTROL & FAULT CONTAINMENT

Detailed subsystem PRD / engineering assurance annex - Up to 5 A per cell can transfer meaningful energy; a stuck switch, wrong source/destination selection or sensor error must not overcharge or overheat a cell.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A09-01	The safety concept shall trace balancing switch topology, current sensing, source/destination authorization and isolation from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	Balancing schematic/SOA review.
SSBMS-A09-02	balancing current regulation and cell-energy bounds shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	5 A capability and boundary-current test.
SSBMS-A09-03	A fault affecting balancing switch/current/temp feedback shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	Stuck-switch/wrong-cell/sensor fault injection.

Verification, margin and future-change rule

Up to 5 A per cell can transfer meaningful energy; a stuck switch, wrong source/destination selection or sensor error must not overcharge or overheat a cell. Balancing fault timing shall capture switch command, current build-up, current/voltage detection, independent inhibit and residual energy after turn-off. Switch technology, current sensor, energy-storage element or balancing state-machine changes require renewed fault-energy and containment verification.

BALANCING THERMAL CROSS-COUPLING & DUTY MANAGEMENT

Detailed subsystem PRD / engineering assurance annex - Balancing heat can bias nearby temperature/pressure measurements and interact with cell aging; duty scheduling must account for local thermal impedance and repeated use.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A10-01	Balancing thermal control shall define the heat-generation model, sensor locations, per-cell duty counters, cooldown rules and the conditions that reduce or suspend balancing when local thermal margin is consumed.	Thermal model/layout review.
SSBMS-A10-02	temperature rise and duty-cycle limits during repeated balancing shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Calorimetric repeated-balancing test.
SSBMS-A10-03	Loss, invalidity or disagreement affecting thermal model or local temperature validity shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Cooling/sensor degradation injection.

Verification, margin and future-change rule

Balancing heat can bias nearby temperature/pressure measurements and interact with cell aging; duty scheduling must account for local thermal impedance and repeated use. Temperature rise and duty-cycle limits during repeated balancing shall be correlated to representative modules. Balancing topology, switch technology, thermal path, sensor placement or per-cell duty-policy changes require renewed electro-thermal verification.

NON-LINEAR KALMAN SOC ESTIMATION & NUMERICAL ROBUSTNESS

Detailed subsystem PRD / engineering assurance annex - SOC estimation combines coulomb counting and model correction while handling pressure/impedance dependence and numeric edge cases.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A11-01	The released estimation function shall configuration-control Kalman state vector, covariance tuning, process/measurement noise and arithmetic bounds, including model version, state variables, calibration set, numerical limits, data quality rules and applicability to each cell/pack variant.	Estimator design/calibration review.
SSBMS-A11-02	SOC convergence, drift and bounded error over validated domain shall meet the released accuracy, convergence and bounded-error criteria over representative temperature, SOC, SOH, pressure, impedance and duty-cycle conditions; extrapolation beyond validated regions shall be explicitly limited.	Reference-profile replay over temperature/pressure.
SSBMS-A11-03	When covariance, current/voltage data or model validity is unavailable, stale or outside the validated model domain, the estimator shall declare reduced confidence and shall not silently increase charge, discharge, balancing or contactor permissiveness.	Dropout/outlier/NaN/overflow injection.

Verification, margin and future-change rule

SOC estimation combines coulomb counting and model correction while handling pressure/impedance dependence and numeric edge cases. The estimator is only authoritative inside the validated data and cell-chemistry domain. Soc convergence, drift and bounded error over validated domain shall be correlated to independent reference data. A cell supplier, model structure, calibration, sensing or numerical implementation change requires model-identification review and regression of every power, balancing or maintenance decision that consumes the estimate.

SOH CAPACITY, RESISTANCE & DEGRADATION ESTIMATION

Detailed subsystem PRD / engineering assurance annex - SOH must distinguish reversible temperature/pressure effects from persistent capacity loss and impedance growth so power limits and service predictions remain defensible.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A12-01	The released estimation function shall configuration-control SOH state variables, capacity/impedance features, aging priors and confidence outputs, including model version, state variables, calibration set, numerical limits, data quality rules and applicability to each cell/pack variant.	SOH model/configuration review.
SSBMS-A12-02	SOH trend accuracy and false-degradation discrimination shall meet the released accuracy, convergence and bounded-error criteria over representative temperature, SOC, SOH, pressure, impedance and duty-cycle conditions; extrapolation beyond validated regions shall be explicitly limited.	Aged-cell dataset and accelerated-aging correlation.
SSBMS-A12-03	When aging history or model-confidence data is unavailable, stale or outside the validated model domain, the estimator shall declare reduced confidence and shall not silently increase charge, discharge, balancing or contactor permissiveness.	History loss/model-domain fault test.

Verification, margin and future-change rule

SOH must distinguish reversible temperature/pressure effects from persistent capacity loss and impedance growth so power limits and service predictions remain defensible. The estimator is only authoritative inside the validated data and cell-chemistry domain. Soh trend accuracy and false-degradation discrimination shall be correlated to independent reference data. A cell supplier, model structure, calibration, sensing or numerical implementation change requires model-identification review and regression of every power, balancing or maintenance decision that consumes the estimate.

CELL IMPEDANCE OBSERVABILITY & EXCITATION QUALITY

Detailed subsystem PRD / engineering assurance annex - Impedance growth is a lifecycle indicator only when the excitation window, SOC, temperature, pressure and current metrology support a meaningful estimate.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A13-01	The released estimation function shall configuration-control impedance-estimation window, excitation quality, filtering and operating-condition qualification, including model version, state variables, calibration set, numerical limits, data quality rules and applicability to each cell/pack variant.	Impedance algorithm review.
SSBMS-A13-02	impedance estimate repeatability and sensitivity to aging shall meet the released accuracy, convergence and bounded-error criteria over representative temperature, SOC, SOH, pressure, impedance and duty-cycle conditions; extrapolation beyond validated regions shall be explicitly limited.	Reference impedance/EIS correlation where applicable.
SSBMS-A13-03	When excitation quality or operating-condition validity is unavailable, stale or outside the validated model domain, the estimator shall declare reduced confidence and shall not silently increase charge, discharge, balancing or contactor permissiveness.	Poor-excitation/noisy-data test.

Verification, margin and future-change rule

Impedance growth is a lifecycle indicator only when the excitation window, SOC, temperature, pressure and current metrology support a meaningful estimate. The estimator is only authoritative inside the validated data and cell-chemistry domain. Impedance estimate repeatability and sensitivity to aging shall be correlated to independent reference data. A cell supplier, model structure, calibration, sensing or numerical implementation change requires model-identification review and regression of every power, balancing or maintenance decision that consumes the estimate.

STALE, FROZEN & PLAUSIBLE-BUT-WRONG SENSOR DATA

Detailed subsystem PRD / engineering assurance annex - A frozen value can remain numerically plausible while hiding a changing cell condition; freshness and cross-domain plausibility are propagated with every safety/performance variable.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A14-01	Measurement-validity handling shall define freshness counters, frozen-value detection, cross-domain plausibility and the propagation of invalid/stale status to every estimator, limit and protection consumer.	Data-validity architecture review.
SSBMS-A14-02	stale/frozen detection and bounded fallback behavior shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	HIL frozen/ramped/disagreement injection.
SSBMS-A14-03	Loss, invalidity or disagreement affecting cell voltage, temperature, pressure or current data validity shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Multi-sensor stale/common-cause test.

Verification, margin and future-change rule

A frozen value can remain numerically plausible while hiding a changing cell condition; freshness and cross-domain plausibility therefore travel with every safety- or performance-relevant variable. Stale/frozen detection and bounded fallback behavior shall be verified with frozen, delayed, repeated and cross-channel-inconsistent data. Signal source, update rate, plausibility model or timeout-policy changes require data-validity regression.

ADC SYNCHRONIZATION, TASK TIMING & WCET RESERVE

Detailed subsystem PRD / engineering assurance annex - Estimation and protection consume measurements sampled at different rates; timing skew and overload must remain inside the assumptions used by algorithms and safety mechanisms.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A15-01	The sampling/timing baseline shall define ADC triggers, synchronization points, ISR priorities, task periods, WCET budgets, interrupt reserve and maximum data age at each safety- or performance-relevant consumer.	Timing architecture/WCET review.
SSBMS-A15-02	timing/jitter/data-age bounds at maximum CPU and communication load shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Instrumented max-load timing test.
SSBMS-A15-03	Loss, invalidity or disagreement affecting task execution or timestamp validity shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Interrupt storm/task stall/timestamp fault injection.

Verification, margin and future-change rule

Estimation and protection consume measurements sampled at different rates; timing skew and overload must remain inside the assumptions used by algorithms and safety mechanisms. Timing, jitter and data-age bounds shall be measured at maximum CPU, diagnostic and communication load. ADC trigger, ISR priority, task period, scheduler, compiler or communication-burst changes reopen WCET and synchronization verification.

WATCHDOG, RESET, BROWNOUT & SAFE-STATE RETENTION

Detailed subsystem PRD / engineering assurance annex - Reset or low-voltage disturbance must not create unintended contactor closure, balancing activity or permissive limits while state and calibration integrity are uncertain.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A16-01	The safety concept shall trace watchdog milestones, reset causes, brownout thresholds and default I/O states from hazard or item-level safety objective to independent detection, reaction, diagnostic coverage and objective verification evidence, with program-specific ASIL allocation configuration controlled.	Reset/power-sequencing review.
SSBMS-A16-02	reset/brownout recovery without permissive transients shall satisfy the released response-time and independence budget under worst credible electrical, thermal, sensor and communication conditions; software shall not be credited where hardware response is required by the FTTI.	Brownout/reset cycling with live HV simulator.
SSBMS-A16-03	A fault affecting watchdog/reset/calibration validity shall not result in an uncontrolled increase of cell voltage, current, temperature, pressure, contactor energy or balancing energy beyond the item safety allocation.	Processor-stall/watchdog/bias fault injection.

Verification, margin and future-change rule

Reset or low-voltage disturbance must not create unintended contactor closure, balancing activity or permissive limits while state and calibration integrity are uncertain. Brownout testing shall sweep supply fall/rise rate, reset-domain timing, NVM/calibration restoration, I/O default states and watchdog recovery while monitoring every energy-control output. PMIC, reset supervisor, watchdog, boot sequence or NVM-recovery changes require repeated permissive-transient regression.

ISOLATED DAISY-CHAIN CRC, RETRY & SEGMENT FAULT CONTAINMENT

Detailed subsystem PRD / engineering assurance annex - A broken module link must be localized and prevented from silently converting downstream cells into apparently healthy stale data.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A17-01	Every released interface for daisy-chain framing, CRC, retry, segmentation and module addressing shall define source, destination, units, encoding, valid range, update rate, freshness/integrity status, timeout and authority; ambiguous ownership is not permitted.	Communication architecture review.
SSBMS-A17-02	latency/error-recovery behavior and downstream fault localization shall remain deterministic and bounded during message loss, duplication, corruption, bus recovery, reset and maximum specified network or daisy-chain loading.	Error-rate/link-break/retry stress test.
SSBMS-A17-03	Invalid, stale or unauthenticated information associated with link freshness, module identity or CRC status shall be treated as non-permissive and shall not override autonomous local voltage, current, thermal, pressure or contactor protection.	Bit error/reordered/missing module injection.

Verification, margin and future-change rule

A broken module link must be localized and prevented from silently converting downstream cells into apparently healthy stale data. Every safety- or performance-relevant signal shall carry ownership, freshness, validity and timeout semantics. Latency/error-recovery behavior and downstream fault localization shall be verified under loss, corruption and overload. Protocol or topology changes require interface, timing and fallback regression; local cell protection may not depend on successful network recovery.

AES-128 KEYS, REPLAY RESISTANCE & MESSAGE AUTHORITY

Detailed subsystem PRD / engineering assurance annex - Encryption alone does not define trust. Key lifecycle, freshness, message origin and authorized diagnostic/service roles determine whether protected monitor data can be trusted.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A18-01	The security architecture shall configuration-control AES-128 key provisioning, storage, rotation and authorized roles, including trust anchors, key ownership, authorized roles, secure state transitions and the boundary between cyber services and local safety protection.	Cybersecurity configuration review.
SSBMS-A18-02	authentication/freshness behavior for modified, replayed or malformed traffic shall reject replayed, modified, malformed or unauthorized data without violating timing, availability or safety requirements for voltage, current, temperature, pressure and contactor supervision.	Replay/modified-frame/negative-auth test.
SSBMS-A18-03	Loss or invalidity of key service, freshness counter or authentication result shall fail closed for privileged access and permissive commands while preserving autonomous local fault detection, isolation and safe-state actions.	Key-loss/counter-reset/security-service fault test.

Verification, margin and future-change rule

Encryption alone does not define trust. Key lifecycle, freshness, message origin and authorized diagnostic/service roles determine whether protected monitor data can be trusted. Cyber controls are subordinate to local safety response. Authentication/freshness behavior for modified, replayed or malformed traffic shall be proven with negative authorization and malformed/replay cases while monitoring timing and safety behavior. Key, loader, diagnostic-role or trust-anchor changes require threat-analysis impact review and targeted regression before release.

CAN FD VEHICLE INTERFACE, LIMIT AUTHORITY & TIMEOUTS

Detailed subsystem PRD / engineering assurance annex - Vehicle controllers consume BMS charge/discharge limits, SOC/SOH and fault status; stale permissive limits must never survive beyond the released timeout policy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A19-01	Every released interface for CAN FD signal ownership, encoding, update rate and command/limit authority shall define source, destination, units, encoding, valid range, update rate, freshness/integrity status, timeout and authority; ambiguous ownership is not permitted.	Vehicle ICD/data-contract review.
SSBMS-A19-02	timeout, bus-off recovery and overload behavior shall remain deterministic and bounded during message loss, duplication, corruption, bus recovery, reset and maximum specified network or daisy-chain loading.	CAN FD conformance and overload test.
SSBMS-A19-03	Invalid, stale or unauthenticated information associated with vehicle command or BMS limit/status freshness shall be treated as non-permissive and shall not override autonomous local voltage, current, thermal, pressure or contactor protection.	Bus-off/stale/duplicate/range fault injection.

Verification, margin and future-change rule

Vehicle controllers consume BMS charge/discharge limits, SOC/SOH and fault status; stale permissive limits must never survive beyond the released timeout policy. Every safety- or performance-relevant signal shall carry ownership, freshness, validity and timeout semantics. Timeout, bus-off recovery and overload behavior shall be verified under loss, corruption and overload. Protocol or topology changes require interface, timing and fallback regression; local cell protection may not depend on successful network recovery.

SECURE BOOT, DIAGNOSTIC ACCESS & CALIBRATION PROTECTION

Detailed subsystem PRD / engineering assurance annex - Service and update paths may alter algorithms or cell limits only under authenticated, mode-controlled conditions; loss of security services cannot delay local safety protection.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A20-01	The security architecture shall configuration-control secure boot, signed software, diagnostic roles, calibration authorization and rollback, including trust anchors, key ownership, authorized roles, secure state transitions and the boundary between cyber services and local safety protection.	Cybersecurity architecture review.
SSBMS-A20-02	modified-image rejection and mode-controlled service behavior shall reject replayed, modified, malformed or unauthorized data without violating timing, availability or safety requirements for voltage, current, temperature, pressure and contactor supervision.	Modified-image and diagnostic negative test.
SSBMS-A20-03	Loss or invalidity of trust anchor, authorization or update validity shall fail closed for privileged access and permissive commands while preserving autonomous local fault detection, isolation and safe-state actions.	Auth-loss/rollback/concurrent safety-fault test.

Verification, margin and future-change rule

Service and update paths may alter algorithms or cell limits only under authenticated, mode-controlled conditions; loss of security services cannot delay local safety protection. Cyber controls are subordinate to local safety response. Modified-image rejection and mode-controlled service behavior shall be proven with negative authorization and malformed/replay cases while monitoring timing and safety behavior. Key, loader, diagnostic-role or trust-anchor changes require threat-analysis impact review and targeted regression before release.

THERMAL MANAGEMENT COORDINATION & POWER LIMIT DERATING

Detailed subsystem PRD / engineering assurance annex - The BMS publishes heat-generation and cell-limit information to the vehicle thermal controller while retaining local authority to reduce charge/discharge/balancing energy when temperature margin is consumed.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A21-01	Thermal coordination shall define cell/AFE/balancing temperature signals, heat-generation data, local derating authority, vehicle-thermal interface ownership and the non-permissive behavior used when thermal data is invalid.	Thermal/vehicle ICD review.
SSBMS-A21-02	derating behavior across hot/cold soak and cooling degradation shall remain bounded across the released cell, pack, temperature, aging and supply-tolerance envelope, including credible single faults and stale or implausible input conditions.	Chamber plus thermal-controller HIL sweep.
SSBMS-A21-03	Loss, invalidity or disagreement affecting thermal data or cooling-interface validity shall transition the BMS to the allocated non-permissive, charge-limited, discharge-limited or isolation state without increasing electrical, thermal or mechanical cell stress.	Cooling loss/sensor invalid/timeout injection.

Verification, margin and future-change rule

The BMS publishes heat-generation and cell-limit information to the vehicle thermal controller while retaining local authority to reduce charge, discharge or balancing energy when temperature margin is consumed. Derating behavior across hot/cold soak and cooling degradation shall be correlated to representative pack hardware. Temperature-sensor placement, thermal model, coolant interface or derating-map changes require renewed thermal-response and limit-authority verification.

PACK PRESSURE LOAD PATH, SENSOR MOUNTING & STRUCTURAL ENERGY

Detailed subsystem PRD / engineering assurance annex - Mechanical preload is part of the cell operating environment; sensing hardware and service actions must not create local stress concentration or invalidate the pressure transfer function.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A22-01	Pack-structure qualification shall define the pressure load path, sensor mounting, compression hardware, structural limits and service stack-up for the released cell format and enclosure configuration.	Mechanical FEA/stack-up review.
SSBMS-A22-02	strain/pressure repeatability after load and environmental cycling shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Instrumented compression/vibration/thermal cycle test.
SSBMS-A22-03	If evidence for mechanical stack-up or sensor-mounting configuration is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Mounting tolerance/service reassembly variation.

Verification, margin and future-change rule

Mechanical preload is part of the cell operating environment; sensing hardware and service actions must not create local stress concentration or invalidate the pressure transfer function. The qualification argument shall bind pack pressure load path, sensor mounting, structural limits and service stack-up to the released configuration, with strain/pressure repeatability demonstrated after load and environmental cycling. A change to cell format, enclosure structure, sensor bracket, compression hardware or service assembly method requires structural/pressure correlation before that variant can be released.

EMC, COMMON-MODE IMMUNITY & MEASUREMENT INTEGRITY

Detailed subsystem PRD / engineering assurance annex - High-voltage switching from inverter, charger and balancing circuits can couple into high-impedance AFE and pressure-sensor wiring; immunity is verified by monitoring safety-relevant data, not only communication uptime.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A23-01	EMC qualification shall define filtering, shielding/grounding, common-mode return paths, measurement-harness routing and the switching environments used to demonstrate cell/current/pressure data integrity.	EMC architecture/layout review.
SSBMS-A23-02	cell/pressure/current measurement integrity during qualified emissions/immunity conditions shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	CISPR/ISO 11452-style program test as applicable.
SSBMS-A23-03	If evidence for sensor plausibility or communication integrity under EMI is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	RF/BCI/transient injection with data logging.

Verification, margin and future-change rule

High-voltage switching from inverter, charger and balancing circuits can couple into high-impedance AFE and pressure-sensor wiring; immunity is verified by monitoring safety-relevant data, not only communication uptime. EMC evidence shall correlate filtering, grounding/shielding, common-mode paths and measurement wiring to the released harness and switching environment. Harness routing, shield termination, grounding, inverter/charger switching spectrum or balancing frequency changes require targeted emissions/immunity regression.

ENVIRONMENTAL, VIBRATION & THERMAL-CYCLE QUALIFICATION

Detailed subsystem PRD / engineering assurance annex - Pack-mounted electronics experience vibration, thermal cycling, humidity/condensation and pressure-stack changes that can shift calibration or insulation before producing a visible functional fault.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A24-01	Environmental qualification shall define the released vibration, thermal-cycle, humidity/condensation, sealing, connector-retention and sequential-stress conditions for each installation variant.	Qualification-plan review.
SSBMS-A24-02	post-stress electrical/pressure calibration and insulation margin shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Sequential environment plus post-test metrology.
SSBMS-A24-03	If evidence for environmental qualification evidence for released pack variant is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Open evidence blocks affected variant release.

Verification, margin and future-change rule

Pack-mounted electronics experience vibration, thermal cycling, humidity/condensation and pressure-stack changes that can shift calibration or insulation before producing a visible functional fault. Environmental qualification shall retain post-stress electrical, pressure-calibration and insulation evidence for the released mounting, sealing and connector configuration. Pack location, sealing system, connector family, mounting stiffness or pressure-stack changes require renewed sequential-stress qualification.

PRESSURE-SENSOR EOL CALIBRATION & MEASUREMENT UNCERTAINTY

Detailed subsystem PRD / engineering assurance annex - Pressure offset logging is valuable only when fixture reference, preload application, temperature condition and measurement uncertainty are controlled and traceable to the serialized unit.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A25-01	Pressure-sensor EOL control shall define the reference chain, preload fixture, calibration temperature, gauge capability, uncertainty budget, offset acceptance and serialized storage of the resulting calibration data.	EOL calibration plan review.
SSBMS-A25-02	calibration repeatability, gauge capability and offset acceptance shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Gauge R&R/golden-unit correlation.
SSBMS-A25-03	If evidence for pressure-calibration evidence and serialized mapping is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Missing/invalid calibration blocks unit release.

Verification, margin and future-change rule

Pressure offset logging is valuable only when fixture reference, preload application, temperature condition and measurement uncertainty are controlled and traceable to the serialized unit. EOL pressure evidence shall include reference traceability, gauge capability, repeatability/reproducibility and golden-unit correlation. Reference sensor, preload tooling, fixture mechanics, temperature compensation or offset-storage changes require fixture recorrelation before production use.

MANUFACTURING PROCESS CAPABILITY & CRITICAL CHARACTERISTICS

Detailed subsystem PRD / engineering assurance annex - Critical assembly characteristics include sensor preload, AFE leakage-sensitive cleanliness, isolation spacing, shunt/busbar assembly, contactor interfaces and balancing-current paths.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A26-01	Manufacturing release shall identify critical characteristics, cleanliness controls, preload/assembly limits, inspection methods, process-capability targets and measurement-system capability for each controlled operation.	Control plan/PFMEA review.
SSBMS-A26-02	process variation against approved trim/assembly limits shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Capability study and golden-unit correlation.
SSBMS-A26-03	If evidence for manufacturing capability and traceability evidence is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Insufficient capability blocks production release.

Verification, margin and future-change rule

Critical assembly characteristics include sensor preload, AFE leakage-sensitive cleanliness, isolation spacing, shunt/busbar assembly, contactor interfaces and balancing-current paths. Manufacturing evidence shall demonstrate process capability against the approved assembly/trim limits without using software trim to hide a physical defect. Material, process, tooling, supplier or inspection-method changes require capability re-evaluation and golden-unit/process correlation before production release.

CELL GENEALOGY, SERIALIZATION & NVM TRACEABILITY

Detailed subsystem PRD / engineering assurance annex - Estimator behavior and lifecycle trends depend on knowing which cell lot, module, sensor calibration and BMS configuration produced the data; genealogy is part of technical evidence, not only logistics.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A27-01	Genealogy control shall link cell lot, module serial number, BMS hardware, sensor calibrations, software build, calibration set and NVM schema so field evidence can be reconstructed after service changes.	Configuration/data-model review.
SSBMS-A27-02	traceability persistence through service replacement and software update shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Serialization/service/update trace test.
SSBMS-A27-03	If evidence for genealogy or configuration identity is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Broken genealogy blocks equivalence claim.

Verification, margin and future-change rule

Estimator behavior and lifecycle trends depend on knowing which cell lot, module, sensor calibration and BMS configuration produced the data; genealogy is part of technical evidence, not only logistics. Traceability shall survive module replacement, BMS replacement, calibration update and software update without breaking the link between field data and the physical configuration. Serialization format, NVM schema, service workflow or backend data-model changes require end-to-end genealogy audit.

FULL-PACK HIL, FAULT INJECTION & DVP CLOSURE

Detailed subsystem PRD / engineering assurance annex - Nominal bench tests do not cover combined sensor, timing, communication, contactor, thermal and pressure conditions. DVP fixes pass/fail limits before execution and retains raw evidence.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A28-01	Full-pack verification release shall define the HIL/pack-HIL matrix, representative plant conditions, combined faults, instrumentation accuracy, raw-data retention and fixed pass/fail criteria before test execution.	DVP readiness review.
SSBMS-A28-02	coverage of operating corners, credible single faults and combined stresses shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Pack-HIL/power-HIL/chamber fault matrix.
SSBMS-A28-03	If evidence for critical DVP evidence or configuration identity is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Open critical evidence keeps verification gate open.

Verification, margin and future-change rule

Nominal bench tests do not cover combined sensor, timing, communication, contactor, thermal and pressure conditions. DVP fixes pass/fail limits before execution and retains raw evidence. The verification matrix shall cover released operating corners, credible single faults and relevant combined stresses using calibrated instrumentation and configuration-controlled plant/load models. Operating-envelope, fault-model, HIL model or test-equipment changes require DVP impact review and repetition of affected rows before equivalence is claimed.

15-YEAR IMPEDANCE GROWTH & PREDICTIVE-MAINTENANCE EVIDENCE

Detailed subsystem PRD / engineering assurance annex - Impedance-growth tracking is meaningful only with trend quality, comparable operating conditions, retained history and service thresholds correlated to cell/pack aging evidence.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A29-01	Lifecycle monitoring release shall define impedance/pressure/thermal counters, normalization conditions, trend-quality metrics, retained history, service thresholds and the evidence used to establish false-call performance.	Lifecycle/field-data governance review.
SSBMS-A29-02	predictive-maintenance confidence and false-call control shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Accelerated aging plus long-history replay.
SSBMS-A29-03	If evidence for lifecycle history or aging-model validity is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Open aging evidence blocks lifecycle equivalence.

Verification, margin and future-change rule

Impedance-growth tracking is meaningful only with trend quality, comparable operating conditions, retained history and service thresholds correlated to cell/pack aging evidence. Lifecycle evidence shall quantify estimator confidence, false-call behavior and service-threshold stability across representative aging trajectories. Cell chemistry, impedance-extraction method, trend filter, retained-history policy or service-threshold changes require renewed aging-model correlation before predictive-maintenance claims are released.

SUPPLIER / SECOND-SOURCE & CELL-CHEMISTRY CHANGE CONTROL

Detailed subsystem PRD / engineering assurance annex - A second-source AFE, pressure sensor, current sensor, balancing switch, contactor or cell family can alter error, thermal, mechanical and estimator assumptions even when nominal interfaces appear compatible.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-A30-01	Supplier/second-source approval shall define PPAP/BOM identity, guaranteed electrical/mechanical limits, cell-chemistry deltas, similarity bounds and the exact analyses/tests that must be repeated before equivalence is claimed.	Supplier-change impact review.
SSBMS-A30-02	delta analysis and regression scope for changed source or cell chemistry shall be demonstrated across the released process, environmental, aging and tolerance envelope using calibrated equipment and predefined pass/fail criteria; waivers require approved impact analysis.	Targeted WCCA/DVP regression on representative lots.
SSBMS-A30-03	If evidence for second-source equivalence or changed-cell evidence is incomplete, outside approved similarity bounds or inconsistent with the released configuration, the affected hardware/BOM/installation variant shall not be released until the evidence gap and regression scope are closed.	Unproven equivalence blocks source approval.

Verification, margin and future-change rule

A second-source AFE, pressure sensor, current sensor, balancing switch, contactor or cell family can alter error, thermal, mechanical and estimator assumptions even when nominal interfaces appear compatible. Supplier approval shall use guaranteed production limits, PPAP/BOM identity, similarity analysis and targeted regression tied to the changed component or chemistry. No alternate source or cell family is production-equivalent until the affected WCCA, estimator, safety, thermal, mechanical and DVP evidence is explicitly closed.

REV A2 RELEASE CLOSURE - ENGINEERING ASSURANCE BASELINE

Requirements, evidence, configuration and future-change closure for the solid-state lithium-metal BMS.

The Rev A2 baseline combines the product-level PRD with detailed subsystem requirements and engineering-assurance annexes. Release requires closure of variant-specific cell and pack parameters, program-specific ISO 26262 allocation, measurement WCCA, estimation-model validity, pressure/mechanical assumptions, cybersecurity configuration, manufacturing capability and objective DVP evidence. No requirement is considered complete only because a related narrative paragraph exists; completion is tied to the specified test, analysis, inspection, timing trace or configuration evidence.

Gate	Evidence required	Release blocker
Architecture / ICD	Released cell/pack limits, state/authority model, sensing, contactor, balancing and pressure interfaces	Unbounded or conflicting ownership
Safety / protection	Closed fault paths, FTTI, diagnostic coverage, independence and item-safety evidence	Open high-severity mechanism
HW / SW / model	WCCA, WCET, calibration bounds, estimator validity and configuration integrity	Unverified derived requirement or model-margin deficit
Verification / manufacturing	DVP with calibrated instrumentation, raw evidence, EOL calibration and process capability	Failed/waived critical test or incapable process
Lifecycle / supplier change	Impedance/pressure aging model, NVM counters, genealogy, service controls and similarity analysis	Unproven lifecycle or source equivalence

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
SSBMS-REL-001	All safety- and performance-relevant TBC/TBD parameters shall be closed, owned and configuration controlled before production release.	Parameter-register closure audit.
SSBMS-REL-002	Every atomic requirement shall have objective pass/fail evidence tied to released hardware, software, calibration, cell/pack variant and test conditions.	Traceability and DVP audit.
SSBMS-REL-003	A failed or waived high-severity test shall block release unless an approved safety/engineering rationale defines containment, residual risk and regression scope.	Release-board evidence review.
SSBMS-REL-004	Supplier, cell chemistry, topology, software, calibration or pack-mechanical changes shall reuse evidence only inside documented similarity bounds.	Change-impact and regression review.
SSBMS-REL-005	Production equivalence shall not be claimed when manufacturing/EOL capability, genealogy, lifecycle model or second-source evidence is incomplete.	Production-release readiness audit.

Verification, margin and future-change rule

Future-proof rule: reuse of existing evidence after a design, cell, supplier, standards or installation change is allowed only inside documented similarity bounds. Any change affecting cell electrical stress, estimator validity, measurement error, balancing energy, contactor/isolation behavior, pressure transfer path, thermal margin, cybersecurity authority or safety independence shall trigger targeted impact analysis and regression DVP before production equivalence is claimed.